Exercise J - A

1. For every integer n, let a_n and b_n be real numbers. Let function $f: R \to R$ be given by

$$f(x)=\begin{cases} a_n+\sin \pi x, & \text{for } x\in[2n,2n+1] \\ b_n+\cos \pi x, & \text{for } x\in(2n-1,2n) \end{cases}$$, for all integers n.

If f is continuous, then which of the following hold(s) for all n : **[JEE-2012]**

प्रत्येक पूर्णांक n के लिये, माना a_n तथा b_n वास्तविक संख्याऐं है। फलन $f: R \to R$ इस प्रकार कि

$$f(x)=\begin{cases} a_n+\sin \pi x, & \text{for } x\in[2n,2n+1] \\ b_n+\cos \pi x, & \text{for } x\in(2n-1,2n) \end{cases}$$, सभी पूर्णांक n के लिये हो।

यदि f सतत् है तो सभी n के लिये निम्न में से कौनसा /कौनसे कथन सत्य है : **[JEE-2012]**

(A) $a_{n-1}-b_{n-1}=0$ (B) $a_n-b_n=1$ (C) $a_n-b_{n+1}=1$ (D) $a_{n-1}-b_n=-1$

Ans. BD

Sol. For f to be continuous :

$f(2n^-)=f(2n^+)$.

$\Rightarrow$ $b_n+\cos 2n\pi = a_n+\sin 2n\pi$

$\Rightarrow$ $b_n+1=a_n$

$\Rightarrow$ $a_n-b_n=1$

($\therefore$ B is correct)

Also $l(x)=\begin{bmatrix} b_n+\cos\pi x & (2n-1,\ 2n) \\ a_n+\sin\pi x & [2n,\ 2n+1] \\ b_{n+1}+\cos\pi x & (2n+1,\ 2n+2) \\ a_n+\sin\pi x & [2n+2,\ 2n+3] \end{bmatrix}$

Again $f((2n+1)^-)=f((2n+1)^+)$

$\Rightarrow$ $a_n=b_{n+1}-1$

$\Rightarrow$ $a_n-b_{n+1}=-1$

$\Rightarrow$ $a_{n-1}-b_n=-1$ ($\therefore$ D is correct)

2. Let $f(x)=\begin{cases} x^2\left|\cos\dfrac{\pi}{x}\right|, & x\neq 0 \\ 0, & x=0 \end{cases}$, $x\in R$, then f is : **[JEE-2012]**

(A) differentiable both at $x=0$ and at $x=2$

(B) differentiable at $x=0$ but not differentiable at $x=2$

(C) not differentiable at $x=0$ but differentiable at $x=2$

(D) differentiable neither at $x=2$ nor at $x=0$

माना $f(x)=\begin{cases} x^2\left|\cos\dfrac{\pi}{x}\right|, & x\neq 0 \\ 0, & x=0 \end{cases}$, $x\in R$,तो f होगा : **[JEE-2012]**

(A) $x = 0$ तथा $x = 2$ पर अवकलनीय है

(B) $x = 0$ पर अवकलनीय परन्तु $x = 2$ पर अवकलनीय नहीं है

(C) $x = 2$ पर अवकलनीय परन्तु $x = 0$ पर अवकलनीय नहीं है

(D) $x = 0$ तथा $x = 2$ दोनों पर अवकलनीय नहीं है

Ans. B

Sol. At $x = 0$

$$\text{R.H.D} = \lim_{h\to 0}\frac{(0+h)-(0)}{h} = \lim_{h\to 0}\frac{h^2\left|\cos\frac{\pi}{h}\right|-0}{h}$$

$$= \lim_{h\to 0} h\left|\cos\frac{\pi}{h}\right| = 0\times\cos(\infty) = 0\times \text{finite} = 0$$

$$\text{LHD} := \lim_{h\to 0}\frac{l(0-h)-l(0)}{-h}$$

$$= \lim_{h\to 0}\frac{h^2\cos\left(\frac{\pi}{-h}\right)-0}{-h} = \lim_{h\to 0} -h\cos\left(\frac{\pi}{h}\right)$$

$$= 0$$

$\because$ LHD = RHD at $x = 0$

$\Rightarrow$ $f(x)$ is differentiable at $x = 0$

At $x = 2$

$$\text{RHD} = \lim_{h\to 0}\frac{l(2+h)-l(2)}{h}$$

$$= \lim_{h\to 0}\frac{(2+h)^2.\cos\left(\frac{\pi}{2+h}\right)-0}{h}$$

$$= 4\lim_{h\to 0}\frac{\cos\left(\frac{\pi}{2+h}\right)}{h}$$

$$= -4\lim_{h\to 0}\frac{\sin\left(\frac{\pi}{2+h}\right).\left(-\frac{\pi}{(2+h)^2}\right)}{1} = \pi$$

LHD : $\lim_{h\to 0}\frac{l(2-h)-l(2)}{-h}$

$$=\lim_{h\to 0}\frac{(2-h)^2\left(-\cos\left(\frac{\pi}{2-h}\right)\right)}{-h}$$

$$=\lim_{h\to 0}\frac{4\left(\sin\frac{\pi}{(2-h)}\right)\left(\frac{\pi}{(2-h)^2}\right)}{-1}=-\pi.$$

LHD $\neq$ RHD at $x = 2$

$\therefore$ Not differentiable at $x = 2$.

3. Let $f_1 : R \to R$, $f_2 : [0, \infty) \to R$, $f_3 : R \to R$ and $f_4 : R \to [0, \infty)$ be defined by

$$f_1(x)=\begin{cases}|x| & \text{if } x<0,\\ e^x & \text{if } x\geq 0;\end{cases}\qquad f_2(x)=x^2;\qquad f_3(x)=\begin{cases}\sin x & \text{if } x<0,\\ x & \text{if } x\geq 0;\end{cases}$$

and $f_4(x)=\begin{cases}f_2(f_1(x)) & \text{if } x<0,\\ f_2(f_1(x))-1 & \text{if } x\geq 0.\end{cases}$: **[JEE(Adv.)2014]**

Column -I	**Column -II**
P. f_4 is	1. onto but not one-one
Q. f_3 is	2. neither continuous nor one-one
R. $f_2 O f_1$ is	3. differentiable but not one-one
S. f_2 is	4. continuous and one-one

(A) P-3; Q-1; R-4; S-2 (B) P-1; Q-3; R-4; S-2

(C) P-3; Q-1; R-2; S-4 (D) P-1; Q-3; R-2; S-4

माना $f_1 : R \to R$, $f_2 : [0, \infty) \to R$, $f_3 : R \to R$ तथा $f_4 : R \to [0, \infty)$ इस प्र कार परिभाषित है

$$f_1(x)=\begin{cases}|x| & \text{if } x<0,\\ e^x & \text{if } x\geq 0;\end{cases}\qquad f_2(x)=x^2;\qquad f_3(x)=\begin{cases}\sin x & \text{if } x<0,\\ x & \text{if } x\geq 0;\end{cases}$$

तथा $f_4(x)=\begin{cases}f_2(f_1(x)) & \text{if } x<0,\\ f_2(f_1(x))-1 & \text{if } x\geq 0.\end{cases}$: **[JEE(Adv.)2014]**

Column -I	**Column -II**
P. f_4 होगा	1. आच्छादक परन्तु एकैकी नहीं
Q. f_3 होगा	2. असतत् तथा बहुएकैकी
R. $f_2 O f_1$ होगा	3. अवकलनीय तथा बहुएकैकी
S. f_2 होगा	4. सतत् तथा एकैकी

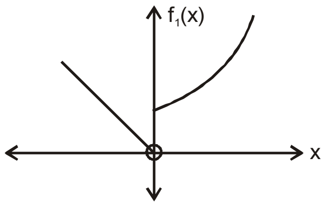
(A) P-3; Q-1; R-4; S-2 (B) P-1; Q-3; R-4; S-2

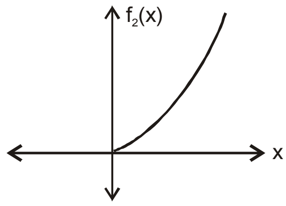
(C) P-3; Q-1; R-2; S-4 (D) P-1; Q-3; R-2; S-4

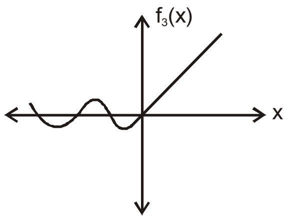
Ans. D

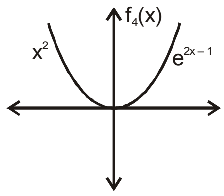
Sol. $f_2(f_1(x)) = (f_1(x))^2 - \begin{bmatrix} x^2 & x<0 \\ e^{2x} & x \geq 0 \end{bmatrix}$

$f_4(x) \begin{bmatrix} x^2 & x<0 \\ e^{2x}-1 & x \geq 0 \end{bmatrix}$









$f_4(x)$ is many-one onto, continuous and non-derivable

$f_3(x)$ is many-one, into, continuous and derivable

$f_2(x)$ is one-one, into, differentiable

$f_4(x)$ बहुएकैकी आच्छादक, सतत् और अवकलनीय नही

$f_3(x)$ बहुएकैकी, अर्न्तक्षेपी, सतत्, अवकलनीय

$f_2(x)$ एकैकी, अर्न्तक्षेपी, अवकलनीय

Hence अत: $R \to 2$

so अत: (D)

$p \to 1, q \to 3, R \to 2, S \to 4$

4. Let $f : R \to R$ and $g : R \to R$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define $h : R \to R$ by

$$h(x) = \begin{cases} \max\{f(x), g(x)\} & \text{if } x \leq 0, \\ \min\{f(x), g(x)\} & \text{if } x > 0 \end{cases}$$

The number of points at which h(x) is not differentiable is : **[JEE(Adv.)2014]**

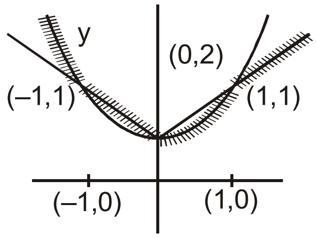
माना $f : R \to R$ तथा $g : R \to R$ क्रमश: दिये गये $f(x) = |x| + 1$ तथा $g(x) = x^2 + 1$. फलन $h : R \to R$ परिभाषित है

$$h(x) = \begin{cases} \max\{f(x), g(x)\} & \text{if } x \leq 0, \\ \min\{f(x), g(x)\} & \text{if } x > 0 \end{cases}$$ बिन्दुओं की संख्या होगी जहाँ h(x) अवकलनीय नहीं है। **[JEE Adv 2014]**

Ans. 3

Sol. $(x) = |x| + 1 = \begin{cases} x+1 & x \geq 0 \\ -x+1 & x < 0 \end{cases}$

$g(x) = x^2 = 1$



Number of Non-differential points 3.
अवलनीय नहीं होने वाले बिन्दुओं की संख्या 3

5. Let $g : R \rightarrow R$ be a differentiable function with $g(0) = 0$, $g'(0) = 0$ and $g'(1) \neq 0$.

Let $f(x) = \begin{cases} \dfrac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$ and $h(x) = e^{|x|}$ for all $x \in R$. Let $(f \circ h)(x)$ denote $f(h(x))$ and

$(h \circ f)(x)$ denote $h(f(x))$. Then which of the following is (are) true ? : **[JEE(Adv.)2015]**

(A) f is differentiable at $x = 0$

(B) h is differentiable at $x = 0$

(C) f o h is differentiable at $x = 0$

(D) h o f is differentiable at $x = 0$

माना कि $g : R \rightarrow R$ एक अवकलनीय फलन है जहाँ कि $g(0) = 0$, $g'(0) = 0$ एवं $g'(1) \neq 0$ है। माना कि

$f(x) = \begin{cases} \dfrac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$ और प्रत्येक $x \in R$ के लिए $h(x) = e^{|x|}$ है। माना कि $(f \circ h)(x)$, $f(h(x))$ को और $(h \circ f)(x)$, $h(f(x))$

को दर्शाते है। तब निम्नलिखित कथनों में से कौन कौन से सही है ? **[JEE(Adv.)2015]**

(A) $x = 0$ पर f अवकलनीय है

(B) $x = 0$ पर h अवकलनीय है

(C) $x = 0$ पर f o h अवकलनीय है

(D) $x = 0$ पर h o f अवकलनीय है

Ans. AD

Sol. $g(0) = 0, \qquad g'(0) = 0 \qquad g'(1) \neq 0$

$$f(x)\begin{cases} g(x) & ; & x > 0 \\ -g(x) & ' & x < 0 \\ 0 & ; & 0 \end{cases} \qquad h(x) = e^{|x|}$$

$$f\left(h(x)\right) = g\left(e^{|x|}\right) \; h\left(f(x)\right) = e^{|g(x)|}$$

$$R\left(f'(0)\right) = \lim_{x \to 10} \frac{g(x) - g(0)}{x - 0} = g'(0) = 0$$

$L\left(f'(0)\right) = \lim_{x\to 10} \frac{-g(x)-g(0)}{x-0} = g'(0) = 0$

$R\left(h'(0)\right) = 1$ & $L\left(h'(0)\right) = -1$ So h(x) is non derivable. at x = 0

अतः h(x), x = 0 पर अवकलनीय नहीं है।

Now $\lim_{x\to 0} \frac{f(h(x))-f(h(0))}{x} = \lim_{x\to 0} \frac{g\left(e^{|x|}\right)-g(1)}{x}$

$R(f'(h(x))) = \lim_{x\to 0^+} \frac{g(e^x)-g(1)}{x} = \lim_{x\to 0^+} \frac{g(e^x)-g(1)}{e^x-1} \frac{e^x-1}{x} = g'(1)$

$L\left(f'(h(x))\right) = -g'(1)$ Hence $f(h(x))$ is non derivable at x = 0

अतः $f(h(x))$, x = 0 पर अवकलनीय नहीं है।

Since x = 0 is repeated root of g(x) So $e^{|g|x||}$ is differetiable at x = 0

चूंकिx = 0, g(x) का पुनरावृत्ति मूल है। अतः $e^{|g|x||}$, x = 0 पर अवकलनीय है।

hence अतः (A), (D)

6. Let $a, b \in R$ and $f: R \to R$ be defined by $f(x) = a\cos(|x^3 - x|) + b|x|\sin(|x^3 + x|)$. Then f is :

[JEE(Adv.)2016]

(A) differentiable at x = 0 if a = 0 and b = 1
(B) differentiable at x = 1 if a = 1 and b = 0
(C) NOT differentaible at x=0 at a=1 and b=0
(D) NOT differentiable at x=1 if a=1 and b=1

माना कि a, b $\in$ R तथा f : R $\to$ R, $f(x) = a\cos(|x^3 - x|) + b|x|\sin(|x^3 + x|)$ से परिभाषित है। तब f :

[JEE(Adv.)2016]

(A) x = 0 पर अवकलनीय है यदि a = 0 और b = 1
(B) x = 1 पर अवकलनीय है यदि a = 1 और b = 0
(C) x = 0 पर अवकलनीय नहीं है यदि a = 1 और b = 0
(D) x = 1 पर अवकलनीय नहीं है यदि a = 1 और b = 1

Ans. AB

Sol. $f(x) = a\cos(x^3 - x) + bx\sin(x^3 + x)$ $\forall\, x \in R$

Which is composition and sum of differentiable functions, therefore f(x) is always continuous and differentiable :

7. Let $f: \left[-\frac{1}{2}, 2\right] \to R$ and $g: \left[-\frac{1}{2}, 2\right] \to R$ be functions defined by $f(x) = [x^2 - 3]$ and $g(x) = |x|\, f(x) + |4x - 7|$ f(x), where [y] denotes the greatest integer less than or equal to y for $y \in R$. Then : **[JEE(Adv.)2016]**

(A) f is discontinuous exactly at three points in $\left[-\frac{1}{2},2\right]$

(B) f is discontinuous exactly at four points in $\left[-\frac{1}{2},2\right]$

(C) g is NOT differentiable exactly at four points in $\left(-\frac{1}{2},2\right)$

(D) g is NOT differentiable exactly at five point in $\left(-\frac{1}{2},2\right)$

माना कि फलन $f:\left[-\frac{1}{2},2\right]\to R$ और $g:\left[-\frac{1}{2},2\right]\to R$, $f(x) = [x^2 - 3]$ और $g(x) = |x|\ f(x) + |4x - 7|\ f(x)$ से परिभाषित हैं, जहाँ $y \in R$ के लिए y से कम या y के बराबर के महत्तम पूर्णांक (greatest integer less than or equal to y) को [y] द्वारा दर्शाया गया है। तब : **[JEE(Adv.)2016]**

(A) $\left[-\frac{1}{2},2\right]$ में f ठीक तीन (exactly three) बिन्दुओं पर असंतत (discontinuous) है

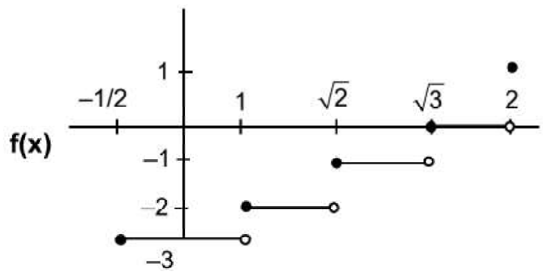
(B) $\left[-\frac{1}{2},2\right]$ में f ठीक चार (exactly four) बिन्दुओं पर असंतत है

(C) $\left(-\frac{1}{2},2\right)$ में g ठीक चार (exactly four) बिन्दुओं पर अवकलनीय (differentiable) नहीं है

(D) $\left(-\frac{1}{2},2\right)$ में g ठीक पाँच (exactly five) बिन्दुओं पर अवकलनीय (differentiable) नहीं है

Ans. BC

Sol.



Clearly from the graph f is discontinuous at four points.

$g(x) = f(x)(|x|+|4x-7|)$

f(x) is non-differentiable at $x = 1, \sqrt{2}, \sqrt{3}$

& $|x|+|4x-7|$ is non-differentiable at $x = 0, \frac{7}{4}$

But $f(x) = 0 \quad \forall x \in \left[\sqrt{3}, 2\right]$

Hence g(x) is non-differentiable at $x = 0, 1, \sqrt{2}, \sqrt{3}$

8. Let [x] be the greatest integer less than or equal to x. Then, at which of the following point(s) the function $f(x) = x\cos(\pi([x]+x))$ is discontinuous? **[JEE(Adv.)2017]**

माना कि x से छोटा या x के समान सबसे बढ़ा पूर्णांक [x] है। तब $f(x) = x\cos(\pi([x]+x))$, निम्न में से किन बिन्दु/बिन्दुओं पर असतत है? **[JEE(Adv.)2017]**

(A) $x = 2$ (B) $x = 1$ (C) $x = -1$ (D) $x = 0$

Ans. ABC

Sol. $f(x) = \cos(\pi x + \pi[x])$

$f(x) \begin{cases} x\cos\pi x & [x] = \text{even} \\ -\cos\pi x & [x] = \text{odd} \end{cases}$

Now check at integer

f(x) is continuous at x = 0

f(x) is disc. at x = 2, 1, –1

9. Let $f_1 : R \to R, f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R, f_3 : \left(-1, e^{\frac{\pi}{2}} - 2\right) \to R$ and $f_4 : R \to R$ defined by :

[JEE(Adv.)2018]

(i) $f_1(x) = \sin\left(\sqrt{1 - e^{-x^2}}\right)$,

(ii) $f_2(x) = \begin{cases} \dfrac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & x = 0 \end{cases}$, where the inverse trigonometric function $\tan^{-1}x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

(iii) $f_3(x) = [\sin(\log_e(x+2))]$, where, for $t \in \mathbb{R}$ [t] denotes the greatest integer less than or equal to t,

(iv) $f_4(x) = \begin{cases} x^2 \sin\left(\dfrac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Column -I	**Column - II**
P. The function f_1 is	1. NOT continuous at x = 0
Q. The function f_2 is	2. continuous at x = 0 and NOT differentiable at x = 0
R. The function f_3 is	3. differentiable at x = 0 and its derivative is NOT continuous at x = 0
S. The function f_4 is	4. differentiable at x = 0 and its derivative is continuous at x = 0

The correct option is: **[JEE(Adv.)2018]**

(A) $P \to 2; Q \to 3; R \to 1; S \to 4$ (B) $P \to 4; Q \to 1; R \to 2; S \to 3$

(C) $P \to 4; Q \to 2; R \to 1; S \to 3$ (D) $P \to 2; Q \to 1; R \to 4; S \to 3$

माना कि फलन $f_1 : R \to R, f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R, f_3 : \left(-1, e^{\frac{\pi}{2}} - 2\right) \to R$ और $f_4 : R \to R$ इस प्रकार परिभाषित हैं कि

[JEE(Adv.)2018]

(i) $f_1(x) = \sin\left(\sqrt{1 - e^{-x^2}}\right)$,

(ii) $f_2(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & x = 0 \end{cases}$, जहाँ प्रतिलोम त्रिकोणमितीय फलन $\tan^{-1}x$, $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में मान धारण करता है,

(iii) $f_3(x) = [\sin(\log_e(x+2))]$, जहाँ $t \in \mathbb{R}$ के लिये, $[t]$, t से छोटा या t के बराबर महत्तम पूर्णांक को दर्शाता है ,

(iv) $f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Column -I	**Column - II**
P. फलन f_1	1. $x = 0$ पर संतत नहीं है
Q. फलन f_2	2. $x = 0$ पर संतत है और $x = 0$ पर अवकलनीय नहीं है
R. फलन f_3	3. $x = 0$ पर अवकलनीय है और $x = 0$ पर इसका अवकलज संतत नहीं है
S. फलन f_4	4. $x = 0$ पर अवकलनीय है और $x = 0$ पर इसका अवकलज संतत है

दिए हुए विकल्पों में से सही विकल्प है : **[JEE(Adv.)2018]**

(A) $P \to 2; Q \to 3; R \to 1; S \to 4$ (B) $P \to 4; Q \to 1; R \to 2; S \to 3$

(C) $P \to 4; Q \to 2; R \to 1; S \to 3$ (D) $P \to 2; Q \to 1; R \to 4; S \to 3$

Ans. D

Sol. (i) $f'_1(0) = \lim_{h \to 0} \frac{\sin\sqrt{1 - e^{-h^2}} - 0}{h} = \lim_{h \to 0} \frac{\sin\sqrt{1 - e^{-h^2}}}{\sqrt{1 - e^{-h^2}}} \times \sqrt{\frac{1 - e^{-h^2}}{h^2}} \times \frac{|h|}{h}$

$= 1 \times 1 \times \frac{|h|}{h} = 1 \times 1 \times \frac{|h|}{h}$

= limit does not exist.

$\Rightarrow$ for option (P), (2) is correct.

(ii) $\lim_{x \to 0} f_2(x) = \lim_{x \to 0} \frac{|\sin x|}{\tan^{-1} x}$

$= \lim_{x \to 0} \frac{|\sin x|}{|x|} \times \frac{x}{\tan^{-1} x} \times \frac{|x|}{x}$

$= \lim_{x \to 0} 1 \times 1 \times \frac{|x|}{x}$

= limit does not exist $\Rightarrow$ for option Q, (1) is correct.

(iii) $= \lim_{x\to 0} f_3(x) = \lim_{x\to 0}[\sin(\log_e(x+2)$ tends to $2]$

now at x tends to zero $(x+2)$ tends to 2

$\Rightarrow$ $\log_e(x+2)$ tends to $ln2$

which is less than 1

$0 < \lim_{x\to 0} \sin(\log_e(x+2)) < \sin 1 \quad \Rightarrow \quad \lim_{x\to 0}[\sin(\log_e(x+2))] = 0$

$f_3(x) = \{0 \quad x \in [-1, e^{\pi/2} - 2)$

$\Rightarrow \quad f'_3(x) = 0 \quad \forall\, x \in (-1, e^{\pi/2} - 2)$

$\Rightarrow \quad f''_3(x) = 0 \quad \forall\, x \in (-1, e^{\pi/2} - 2)$

Hence for (R), (4) is correct.

(iv) $\lim_{x\to 0} f_4(x) = \lim_{x\to 0}\left(x^2 \sin\frac{1}{x}\right) = \lim_{x\to 0} x^2\left(\sin\frac{1}{x}\right) = 0$

$$f'_4(0) = \lim_{h\to 0}\frac{h^2\sin\left(\frac{1}{x}\right) - 0}{x} = \lim_{h\to 0} h\sin\left(\frac{1}{x}\right) = 0$$

$$f'_4(x) = -\cos\frac{1}{x} + x\sin\frac{1}{x},\, x \neq 0$$

$$f''_4(0) = \frac{-\cos\frac{1}{h} + h\sin\frac{1}{h} - 0}{h} \quad \Rightarrow \quad \text{does not exist}$$

hence for (S), (3) is correct.

10. Let f : R$\rightarrow$ R be a function. We say that f has

PROPERTY 1 if $\lim_{h\to 0}\frac{f(h) - f(0)}{\sqrt{|h|}}$ exists and is finite and

PROPERTY 2 if $\lim_{h\to 0}\frac{f(h) - f(0)}{h^2}$ exists and is finite and

Then which of the following options is/are correct ? **[JEE(Adv.)2019]**

(A) $f(x) = |x|$ has PROPERTY 1 (B) $f(x) = x|x|$ has PROPERTY 2

(C) $f(x) = \sin x$ has PROPERTY 2 (D) $f(x) = x^{2/3}$ has PROPERTY 1

माना कि f : R$\rightarrow$ R एक फलन है। हम कहते हैं कि f में

गुण 1 (PROPERTY 1) है यदि $\lim\limits_{h\to0}\dfrac{f(h)-f(0)}{\sqrt{|h|}}$ का अस्तित्व (exists) है और वह परिमित (finite) है,

गुण 2 (PROPERTY 2) है यदि $\lim\limits_{h\to0}\dfrac{f(h)-f(0)}{h^2}$ का अस्तित्व (exists) है और वह परिमित (finite) है।

तब निम्न में से कौन सा (से) विकल्प सही है (हैं) ? **[JEE(Adv.)2019]**

(A) $f(x) = |x|$ में गुण 1 है (B) $f(x) = x|x|$ में गुण 2 है

(C) $f(x) = \sin x$ में गुण 2 है (D) $f(x) = x^{2/3}$ में गुण 1 है

Ans. AD

Sol. Option (1) $f(x) = |x|$ $f(0) = 0$

Property -1

$\lim\limits_{h\to0}\dfrac{|h|}{\sqrt{|h|}} = \lim\limits_{h\to0}|h|^{1/2} = 0$ limit exists

Option (2) $f(x) = x|x|$ $f(0) = 0$

Property -2

$\lim\limits_{h\to0}\dfrac{h|h|-0}{h^2} = \lim\limits_{h\to0}|\dfrac{|h|}{h}$

LHL = –1

RHL = 1

LHL $\neq$ RHL

$\Rightarrow$ Does not exist

Option (3) $f(x) = \sin x$ $f(0) = 0$

Property -2

$\lim\limits_{h\to0}\dfrac{\sin h}{h^2} = \lim\limits_{h\to0}\left(\dfrac{\sin h}{h}.\dfrac{1}{h}\right)$

$\Rightarrow$ Does not exist

Option (4) $f(x) = x^{2/3}$

$= (x^2)^{1/3} = |x|^{2/3} f(0) = 0$

Property -1

$\lim\limits_{h\to0}\dfrac{|h|^{2/3}-0}{\sqrt{|h|}} = \lim\limits_{h\to0}|h|^{2/3-1/2} = \lim\limits_{h\to0}|h|^{1/6} = 0$ limit existsi

11. Let the function $f: R \to R$ be defined by $f(x) = x^3 - x^2 + (x-1)\sin x$ and let $g: R \to R$ be an arbitrary function. Let $fg: R \to R$ be the product function defined by $(fg)(x) = f(x)\, g(x)$. Then which of the following statements is/are TRUE? **[JEE(Adv.)2020]**

(A) If g is continuous at x =1, then fg is differentiable at x =1

(B) If fg is differentiable at x = 1, then g is continuous at x =1

(C) If g is differentiable at x =1, then f g is differentiable at x =1

(D) If f g is differentiable at x =1, then g is differentiable at x =1

माना कि फलन $f : R \to R$ को $f(x) = x^3 - x^2 + (x - 1)\sin x$ द्वारा परिभाषित किया जाता है, एवं माना कि $g : R \to R$ एक स्वेच्छ फलन है। माना कि $f g : R \to R$ गुणन फलन है जो कि $(f g)(x) = f(x)\,g(x)$ के द्वारा परिभाषित है। तब निम्न में से कौन सा (से) कथन सही है (हैं) ? **[JEE(Adv.)2020]**

(A) यदि x =1 पर g संतत है, तब x =1 पर f g अवकलनीय है

(B) यदि x = 1 पर f g अवकलनीय है, तब x =1 पर g संतत है

(C) यदि x =1 पर g अवकलनीय है, तब x =1 पर f g अवकलनीय है

(D) यदि x =1 पर f g अवकलनीय है, तब x =1 पर g अवकलनीय है

Ans. AC

Sol. $f(x) = x^3 - x^2 + (x - 1)\sin(x)$

$f(1) = 0$

f is continous & differentiable

$f'(1) = 1 + \sin 1$

Let $h(x) = f(x)\,g(x)$, $h(1) = 0$

Differentiability at x = 1

$$h'(1^+) = \lim_{h\to\infty} \frac{f(1+h)\,g(1+h) - 0}{h}$$

$= f'(1)\,g(1^+)$

$$h'(1^-) = \lim_{h\to\infty} \frac{f(1-h)\,g(1-h) - 0}{-h}$$

$= f'(1)\,g(1^-)$

(A) If g is continuous at x = 1

$\Rightarrow g(1^+) = g(1) = g(1^-)$

$\Rightarrow h'(1^+) = h'(1^-)$

(B) If $h'(1^+) = h'(1^-)$

$\Rightarrow g(1^+) = g(1^-)$

but g(1) can be different

(C) If g is differentiable at x = 1

$\Rightarrow$ g is continuous at x = 1

from A we can say that C is true

(B) is false so (D) is also false

12. Let $f : R \to R$ and $g : R \to R$ be functions satisfying

$f(x+y) = f(x) + f(y) + f(x)f(y)$ and $f(x) = xg(x)$

for all x, y $\in$ R. If $\lim_{x\to 0} g(x) = 1$, then which of the following statements is/are TRUE ? **[JEE(Adv.)2020]**

(A) f is differentiable at every $x \in R$ (B) If $g(0) = 1$, then g is differentiable at every $x \in R$

(C) The derivative $f'(1)$ is equal to 1 (D) The derivative $f'(0)$ is equal to 1

माना फलन $f: R \to R$ और $g: R \to R$ ऐसे फलन हैं जो कि सभी $x, y \in R$ के लिये

$f(x+y) = f(x) + f(y) + f(x)f(y)$ एवं $f(x) = xg(x)$

को सन्तुष्ट करते हैं। यदि $\lim_{x\to 0} g(x) = 1$ है, तब निम्न में से कौन सा (से) कथन सही है (हैं) ? **[JEE(Adv.)2020]**

(A) f प्रत्येक $x \in R$ पर अवकलनीय है

(B) यदि $g(0) = 1$ है, तब g प्रत्येक $x \in R$ पर अवकलनीय है

(C) अवकलज $f'(1)$ का मान 1 के बराबर है

(D) अवकलज $f'(0)$ का मान 1 के बराबर है

Ans. ABD

Sol. $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h\to 0} \frac{f(x+h) - f(x+0)}{h}$

$= \lim_{h\to 0} \frac{f(x) + f(h) + f(x)f(h) - f(x) - f(0) - f(x)f(0)}{h}$

$f'(x) = \lim_{h\to 0} \frac{f(h) - f(0) + f(x)\left(f(h) - f(0)\right)}{h}$

$f'(x) = \lim_{h\to 0} \left(\frac{f(h) - f(0)}{h}\right)\left(f(x) + 1\right)$

$f'(x) = \lim_{h\to 0} \left(\frac{hg(h) - (0)g(0)}{h}\right)\left(f(x) + 1\right)$

$f'(x) = \lim_{h\to 0} g(h).(f(x) + 1)$

$f'(x) = f(x) + 1$

$\int \frac{f'(x)}{f(x)+1} dx = \int 1 dx$

$\ln(f(x) + 1) = x + c$ $\{\therefore f(0) = 0\}$

$c = 0$

$f(x) = e^x - 1$

f is differentiable at every $x \in R$

$g(x) = \frac{f(x)}{x} = \frac{e^x - 1}{x}$

Check at $x = 0$

$$g'(0) = \lim_{h \to 0} \frac{g(0+h) - g(0)}{h}$$

$$\lim_{h \to 0} \frac{\frac{e^h - 1}{h} - 1}{h}$$

$$\lim_{h \to 0} \frac{1 + h + \frac{h^2}{2!} + \ldots\ldots\ldots\ldots - 1 - h}{h}$$

$$g'(0) = \frac{1}{2}$$

If $g(0) = 1$ then g is differentiable at every $x \in R$

$f'(1) = f(1) + 1 = e - 1 + 1 = e$

$f'(0) = f(0) + 1 = 1$

13. Let the functions $f : (-1, 1) \to R$ and $g : (-1, 1) \to (-1, 1)$ be defined by
$f(x) = |2x - 1| + |2x + 1|$ and $g(x) = x - [x]$,
where [x] denotes the greatest integer less than or equal to x. Let $f \circ g : (-1, 1) \to R$ be the composite function defined by $(f \circ g)(x) = f(g(x))$. Suppose c is the number of points in the interval (–1, 1) at which f o g is NOT continuous, and suppose d is the number of points in the interval (–1, 1) at which f o g is NOT differentiable. Then the value of c + d is ———. **[JEE(Adv.)2020]**

माना कि फलनों $f : (-1, 1) \to R$ एवं $g : (-1, 1) \to (-1, 1)$ को $f(x) = |2x - 1| + |2x + 1|$ एवं $g(x) = x - [x]$ से परिभाषित किया जाता है, जहाँ [x] उस महत्तम पूर्णांक को दर्शाता है जो x से कम या x के बराबर है। माना कि $f \circ g : (-1, 1) \to R$ संयुक्त फलन है जो कि $(f \circ g)(x) = f(g(x))$ द्वारा परिभाषित है। मान लीजिये कि c, अन्तराल (–1, 1) में उन बिंदुओं की संख्या है, जिन पर f o g संतत नहीं है, एवं d, अन्तराल (–1, 1) में उन बिंदुओं की संख्या है, जिन पर f o g अवकलनीय नहीं है। तब c + d का मान है ———

[JEE(Adv.)2020]

Ans. 4

Sol. $f(x) = |2x - 1| + |2x + 1|\, g(x) = x - [x]$

$g(x) = [x] + \{x\} - [x]$

$g(x) = \{x\}$

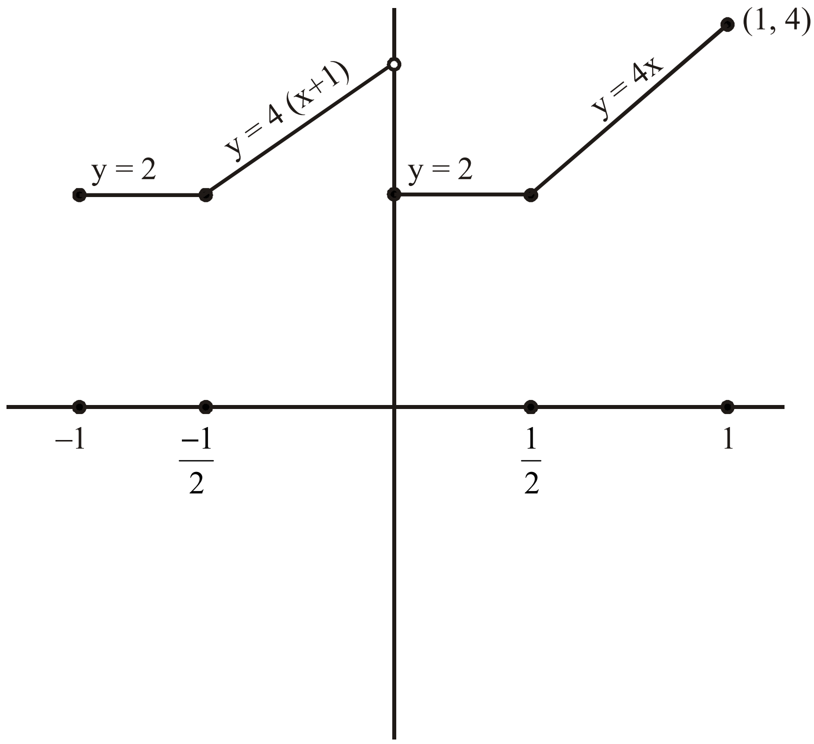
$$f(x) = \begin{bmatrix} 4x & : x \geq \frac{1}{2} \\ 2 & : \frac{-1}{2} < x < \frac{1}{2} \\ -4x & : x \leq \frac{-1}{2} \end{bmatrix}$$

$$f\left(g(x)\right)=\begin{cases} 4\{x\} & :\{x\}\ge \dfrac{1}{2} \\ 2 & :\dfrac{-1}{2}<\{x\}<\dfrac{1}{2} \\ -4\{x\} & :\{x\}\le \dfrac{-1}{2} \end{cases}$$

$$f\left(g(x)\right)=\begin{cases} 4\{x\} & :\{x\}\ge \dfrac{1}{2} \\ 2 & :0\le\{x\}<\dfrac{1}{2} \end{cases}$$

$$f\left(g(x)\right)=\begin{cases} 4\{x\} & :x\in\left[\dfrac{1}{2},1\right)\cup\left[\dfrac{-1}{2},0\right) \\ 2 & :\left[0,\dfrac{1}{2}\right)\cup\left(\dfrac{-1}{2},-1\right) \end{cases}$$

$$f\left(g(x)\right)=\begin{cases} 4x & :x\in\left[\dfrac{1}{2},1\right) \\ 4\left(x+1\right) & :x\in\left[\dfrac{-1}{2},0\right) \\ 2 & :x\in\left[0,\dfrac{1}{2}\right)\cup\left(\dfrac{-1}{2},-1\right) \end{cases}$$



Not continuous at x = 0

Not differetiable at $x = \pm\frac{1}{2}, 0$

c = 1
d = 3
c + d = 1 + 3 = 4

14. Let $S = (0,1) \cup (1,2) \cup (3,4)$ and $T = \{0,1,2,3\}$. Then which of the following statements is(are) true?

[JEE(Adv.)2023]

(A) There are infinitely many functions from S to T

(B) There are infinitely many strictly increasing functions from S to T

(C) The number of continuous functions from S to T is at most 120

(D) Every continuous function from S to T is differentiable

माना कि $S = (0,1) \cup (1,2) \cup (3,4)$ एवं $T = \{0,1,2,3\}$ है। तब निम्न में से कौनसा(से) कथन सत्स है(हैं)?

[JEE(Adv.)2023]

(A) S से T तक अनन्ततः अनेक (Infinitely many) फलन हैं

(B) S से T तक अनंततः अनेक (infinitely many) निरंतर वर्धमान (strictly increasing) फलन हैं

(C) S से T तक संतत (continuous) फलनों की संख्या 120 या उससे कम है

(D) S से T तक प्रत्येक संतत फलन अवकलनीय (differentiable) है

Ans. ACD

Sol. $S=(0,1)\cup(1,2)\cup(3,4)$

$T = \{0,1,2,3\}$

Number of functions

Each element of S have 4 choice

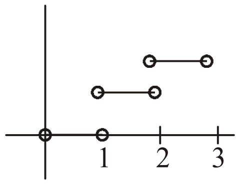
Let n be the number of element in set S.

Number of function $= 4^n$

Here $n \to \infty$

$\Rightarrow$ Option (A) is correct.

Option (B) is incorrect (obvious)

(C) 

For continuous function

Each interval will have 4 choices

$\Rightarrow$ Number of continuous functions

$= 4\times4\times4 = 64$

$\Rightarrow$ Option (C) is correct.

(D) Every continuous function is piecewise constant functions

$\Rightarrow$ Differentiable.

Option (D) is correct.

15. Let $f:(0,1)\to R$ be the function defined as $f(x)=[4x]\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right)$, where [x] denotes the greatest integer less than or equal to x. Then which of the following statements is (are) true ? **[JEE(Adv.)2023]**

(A) The function f is discontinuous exactly at one point in (0, 1)

(B) There is exactly one point in (0, 1) at which the function f is continuous but **NOT** differentiable

(C) The function f is **NOT** differentiable at more than three points in (0, 1)

(D) The minimum value of the function f is $-\frac{1}{512}$

माना कि फलन $f:(0,1)\to R$ इस प्रकार परिभाषित है कि $f(x)=[4x]\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right)$, जहाँ x से कम या x के बराबर महत्तम पूर्णांक (greatest integer) को निरूपित करता है। तब निम्न में से कौनसा (से) कथन सत्य है (हैं) ?

[JEE(Adv.)2023]

(A) फलन f अन्तराल (0, 1) में केवल एक बिन्दु पर असंतत (discontinuous) है

(B) अन्तराल (0, 1) में केवल एक बिन्दु है जिस पर फलन f संतत है किन्तु अवकलनीय नहीं (continuous but not differentiable) है

(C) अन्तराल (0, 1) के तीन से अधिक बिन्दुओं पर फलन f अवकलनीय नहीं (not differentiable) है

(D) फलन f का न्यूनतम मान (minimum value) $-\frac{1}{512}$ है

Ans. AB

Sol. $f:(0,1)\to R$

$$f(x)=[4x]\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right) \qquad \Rightarrow \text{ Critical point } =\frac{1}{4},\frac{1}{2},\frac{3}{4}$$

Discontinuity at $x=\frac{3}{4}$

Continuous and differentiable at $x=\frac{1}{4}$

Continuous but non-differentiable at $x=\frac{1}{2}$

$\text{LHD}\left(\text{at } x=\frac{1}{4}\right)$ $\qquad$ $\text{RHD}\left(\text{at } x=\frac{1}{4}\right)$

$$\lim_{h\to0^+}\frac{0-0}{-h}=0 \qquad \lim_{h\to0^+}\frac{h^2\left(-\frac{1}{2}+h\right)}{h}=0$$

$\text{LHD}\left(\text{at } x=\frac{1}{2}\right)$ $\qquad$ $\text{RHD}\left(\text{at } x=\frac{1}{2}\right)$

$$\lim_{h\to0^+}\frac{\left(\frac{1}{4}-h\right)^2(-h)-0}{-h}-\frac{1}{16} \qquad \lim_{h\to0^+}\frac{2\left(\frac{1}{4}+h\right)^2 h-0}{h}-\frac{1}{8}$$

Minimum -ve value will exist between $\frac{1}{4}$ and $\frac{1}{2}$

$$f(x)=\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right) \qquad \frac{1}{4}\le x\le\frac{1}{2}$$

$$f'(x)=\left(x-\frac{1}{4}\right)\left(3x-\frac{5}{4}\right) \qquad \Rightarrow \qquad \text{minima at } x=\frac{5}{12}$$

$$f\left(\frac{5}{12}\right)=\frac{1}{36}\times\frac{-1}{12} \quad =\frac{-1}{432}$$

16. Let $f : R \to R$ and $g : R \to R$ be functions defined by $f(x)=\begin{cases} x|x|\sin\left(\frac{1}{x}\right) & ,x\neq 0 \\ 0, & x=0, \end{cases}$ and

$g(x)=\begin{cases} 1-2x, & 0\le x\le \frac{1}{2}, \\ 0, & \text{otherwise}, \end{cases}$. Let $a,b,c,d \in R$. Define the function $h : R \to R$ by

$$h(x)=a\,f(x)+b\left(g(x)+g\left(\frac{1}{2}-x\right)\right)+c\left(x-g(x)\right)+dg(x), x\in R.$$

Match each entry in **List-I** to the correct entery in **List -II**.

List -I	**List -II**
(P) If $a = 0$, $b = 1$, $c = 0$, and $d = 0$, then	(1) h is one-one
(Q) If $a = 1$, $b = 0$, $c = 0$, and $d = 0$, then	(2) h is onto
(R) If $a = 0$, $b = 0$, $c = 1$, and $d = 0$, then	(3) h is differentiable on R
(S) If $a = 0$, $b = 0$, $c = 0$, and $d = 1$, then	(4) the range of h is $[0, 1]$
	(5) the range of h is $\{0, 1\}$

The correct option is : **[JEE(Adv.)2024]**

माना कि फलन $f: R \to R$ और $g : R \to R$, $f(x)=\begin{cases} x|x|\sin\left(\frac{1}{x}\right) & ,x\neq 0 \\ 0, & x=0, \end{cases}$ और $g(x)=\begin{cases} 1-2x, & 0\le x\le \frac{1}{2}, \\ 0, & \text{otherwise}, \end{cases}$ द्वारा परिभाषित हैं। माना कि $a,b,c,d \in R$ हैं। फलन $h : R \to R$ को

$$h(x)=a\,f(x)+b\left(g(x)+g\left(\frac{1}{2}-x\right)\right)+c\left(x-g(x)\right)+dg(x), x\in R$$ द्वारा परिभाषित कीजिये।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कीजिए।

सूची-I	**सूची-II**
(P) यदि $a = 0$, $b = 1$, $c = 0$, और $d = 0$ है, तब	(1) h एकैकी है।
(Q) यदि $a = 1$, $b = 0$, $c = 0$, और $d = 0$ है, तब	(2) h आच्छादी है।
(R) यदि $a = 0$, $b = 0$, $c = 1$, और $d = 0$ है, तब	(3) h, R पर अवकलनीय है।
(S) यदि $a = 0$, $b = 0$, $c = 0$, और $d = 1$ है, तब	(4) h का परिसर $[0, 1]$ है।
	(5) h का परिसर $\{0, 1\}$ है।

सही विकल्प है : **[JEE(Adv.)2024]**

(A) (P) $\to$ (4); (Q) $\to$ (3); (R) $\to$ (1); (S) $\to$ (2)

(B) (P) $\to$ (5); (Q) $\to$ (2); (R) $\to$ (4); (S) $\to$ (3)

(C) (P) $\to$ (5); (Q) $\to$ (3); (R) $\to$ (2); (S) $\to$ (4)

(D) (P) $\to$ (4); (Q) $\to$ (2); (R) $\to$ (1); (S) $\to$ (3)

Ans. C

Sol. (P) $h(x)=g(x)+g\left(\frac{1}{2}-x\right)$

$$g(x)=\begin{cases}1-2x & ; \quad 0\le x\le \frac{1}{2}\\ 0 & ; \quad \text{otherwise}\end{cases}$$

$$g\left(\frac{1}{2}-x\right)=\begin{cases}2x & : \quad 0\le x<\frac{1}{2}\\ 0 & : \quad \text{otherwise}\end{cases}$$

$$h(x)=\begin{cases}1 & : \quad 0\le x<\frac{1}{2}\\ 0 & : \quad \text{otherwise}\end{cases}$$

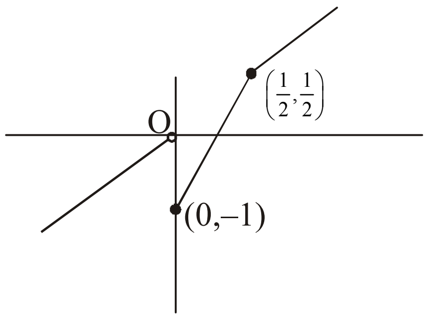
hence, correct option is (5).

(Q) $h(x)=\begin{cases}x|x|\sin\frac{1}{x}; & x\ne 0\\ 0; & \text{otherwise}\end{cases}$

$$h(x)=\begin{cases}x^2\sin\frac{1}{x} & : \quad x>0\\ -x^2\sin\frac{1}{x} & : \quad x<0\\ 0 & : \quad x=0\end{cases}$$

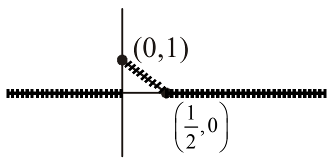
Clearly, function is differentiable everywhere. Hence correct option is (3).

(R) $h(x)=\begin{cases}3x-1 & : \quad 0\le x\le\frac{1}{2}\\ x & : \quad \text{otherwise}\end{cases}$



Clearly onto. Hence, correct option is (2).

(S) $g(x)=\begin{cases}1-2x & : \quad 0\le x\le \dfrac{1}{2}\\ 0 & : \quad \text{otherwise}\end{cases}$



Clearly, range is [0,1].

Hence, correct option is (4).

Hence, overall answer is (C).

17. Let R denote the set of all real numbers. Define the function $f: R \to R$ by :

$$f(x)=\begin{cases}2-2x^2-x^2\sin\dfrac{1}{x} & \text{if} \quad x\ne 0,\\ 2 & \text{if} \quad x=0.\end{cases}$$

Then which of the following statements is TRUE ? **[JEE(Adv.)2025]**

(A) The function f is **NOT** differentiable at $x = 0$

(B) There is a positive real number δ, such that f is a decreasing function on the interval $(0, \delta)$

(C) For any positive real number δ, the function f is **NOT** an increasing function on the interval $(-\delta, 0)$

(D) $x = 0$ is a point of local minima of f

मान लीजिए कि R सभी वास्तविक संख्याओं (real numbers) के समुच्चय (set) को दर्शाता है।

फलन (function) $f: R \to R$ को :

$$f(x)=\begin{cases}2-2x^2-x^2\sin\dfrac{1}{x} & \text{यदि} \quad x\ne 0,\\ 2 & \text{यदि} \quad x=0.\end{cases}$$

द्वारा परिभाषित कीजिये। तब निम्नलिखित कथनों में से कौन सा एक कथन सत्य है ? **[JEE(Adv.)2025]**

(A) $x = 0$ पर, फलन f अवकलनीय **नहीं** है

(B) एक धनात्मक वास्तविक संख्या (positive real number) δ इस प्रकार है कि अंतराल $(0, \delta)$ में फलन f एक ह्रासमान फलन है

(C) किसी भी धनात्मक वास्तविक संख्या δ के लिए अंतराल $(-\delta, 0)$ में फलन f एक वर्धमान फलन **नहीं** है

(D) $x = 0$, फलन f का एक स्थानीय निम्नतम का बिन्दु है

Ans. C

Ans. (By Martix B)

Sol. Calculate LHD & RHD at $x = 0$

$$f'(0^+) = \lim_{h\to 0}\frac{f(0+h)-f(0)}{h}$$

$$= \lim_{h\to 0}\frac{-2h^2 - h^2\sin\left(\frac{1}{h}\right)}{h}$$

$$= \lim_{h\to 0}\left(-h\left(2+\sin\left(\frac{1}{h}\right)\right)\right)$$

$f'(0^+) = 0$

$$f'(0^-) = \lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}$$

$$= \frac{-2h^2 + h^2\sin\left(\frac{1}{h}\right)}{-h}$$

$$= \lim_{h\to 0} h\left(2-\sin\left(\frac{1}{h}\right)\right) = 0$$

In nbd of $x = 0$

$f'(0^+) < 0$ $\quad$ $f'(0^-) > 0$

$\Rightarrow x = 0$ is point of local maxima

f is increasing in left nbd of $x = 0$

f is decreasing in right nbd of $x = 0$

Exercise C - I

1. Which statement is false :

(A) $f(x) = \underset{n\to\infty}{\text{Limit}} \dfrac{1}{1+n\sin^2 \pi x}$ is continuous at x = 1.

(B) The function $f(x) = 2^{-2^{1/(1-x)}}$ if $x \neq 1$ & $f(1) = 1$ is not continuous at x = 1.

(C) There exists a continuous function f: [0, 1] onto [0, 10], but there exists no continuous function g : [0, 1] onto (0, 10).

(D) If $f(x)$ is continuous in [0, 1] & f(x) = 1 for all rational numbers in [0, 1] then $f\left(1/\sqrt{2}\right)$ equal to 1.

(E) If $f(x) = \begin{cases} \dfrac{\cos \pi x + \sin(\pi x/2)}{(x-1)(3x^2-2x-1)} & \text{if } x \neq 1 \\ k & \text{if } x = 1 \end{cases}$ is continuous at x = 1, then the value of k is $\dfrac{3\pi^2}{32}$.

निम्नलिखित में से असत्य कथन है :

(A) $f(x) = \underset{n\to\infty}{\text{Limit}} \dfrac{1}{1+n\sin^2 \pi x}$, x = 1 पर सतत् है।

(B) फलन $f(x) = 2^{-2^{1/(1-x)}}$, $x \neq 1$ और f(1) = 1, x = 1 पर सतत् नहीं है।

(C) एक सतत् आच्छादक फलन f: [0, 1] $\to$ [0, 10] में हो सकता है लेकिन g : [0, 1]$\to$ (0, 10) सतत् आच्छादक नही हो सकता।

(D) फलन f (x), [0 , 1] के अन्तराल में सतत् है तथा इसी अन्तराल में प्रत्येक परिमेय संख्या x के लिये f(x) = 1 तो $f\left(1/\sqrt{2}\right) = 1$.

(E) फलन $f(x) = \begin{cases} \dfrac{\cos \pi x + \sin(\pi x/2)}{(x-1)(3x^2-2x-1)} & \text{if } x \neq 1 \\ k & \text{if } x = 1 \end{cases}$, x = 1 पर सतत है तो k का मान $\dfrac{3\pi^2}{32}$ होगा।

Ans. A

Sol. (B) $f(x) = \begin{cases} 2^{-2^{\frac{1}{1-x}}} & x \neq 1 \\ 1 & x = 1 \end{cases}$

at x = 1

$$\text{RHL} = \lim_{h\to 0} 2^{-2^{-\frac{1}{h}}} = 1$$

$$\text{LHL} = \lim_{h\to 0} 2^{-2^{\frac{1}{h}}} = 0$$

$\therefore \lim\limits_{x\to 1} f(x)$ does not exists.

$\Rightarrow$ f(x) is not continuous at x = 1.

(C) $f:[0,1]\xrightarrow[\text{onto}]{\text{continuous}}[0,10] \Rightarrow$ let $f(x) = 10x^2$

$g:[0,1]\xrightarrow[\text{onto}]{}(0,10)$

$\Rightarrow g:[0,1]\xrightarrow[\text{onto}]{\text{continuous}}(0,10)$

(D) $f(x)=\begin{cases} 1 & x \in Q \cap [0,1] \\ b(\text{let}) & x \in Q^C \cap [0,1] \end{cases}$

Given f(x) is continuous $\forall x \in [0,1]$

$\Rightarrow b = 1$

$\Rightarrow f(x) = 1 \qquad \forall x \in [0,1]$

$\Rightarrow f\left(\frac{1}{\sqrt{2}}\right)=1$

(E) $f(x)=\begin{cases} \dfrac{\cos \pi x+\sin\left(\frac{\pi x}{2}\right)}{(x-1)(3x^2-2x-1)} & \text{if} \quad x \neq 1 \\ k & \text{if} \quad x = 1 \end{cases}$

$$\Rightarrow \lim_{x\to 1}\frac{\cos \pi x+\sin\frac{\pi x}{2}}{(x-1)(3x^2-2x-1)}$$

$$=\lim_{x\to 1}\frac{\sin\frac{\pi}{2}x+\sin\left(\frac{\pi}{2}-\pi x\right)}{(x-1)^2(3x+1)}$$

$$=\frac{1}{4}\lim_{x\to 1}\frac{2\sin\left(\dfrac{\frac{\pi x}{2}+\frac{\pi}{2}-\pi x}{2}\right)\cos\left(\dfrac{\frac{\pi x}{2}-\frac{\pi}{2}+\pi x}{2}\right)}{(x-1)^2}$$

$$=\frac{1}{2}\lim_{x\to 1}\frac{\sin\left(\frac{\pi}{4}-\frac{\pi x}{4}\right)\cos\left(\frac{3\pi x}{4}-\frac{\pi}{4}\right)}{(x-1)^2}$$

$$=\frac{1}{2}\cdot\left(-\frac{\pi}{4}\right)\lim_{x\to1}\frac{\sin\left(\frac{\pi}{2}-\frac{3\pi x}{4}+\frac{\pi}{4}\right)}{(x-1)}$$

$$=+\frac{\pi}{8}\lim_{x\to1}\cdot\frac{3\pi}{4}\frac{\sin\left(\frac{3\pi}{4}(x-1)\right)}{\frac{3\pi}{4}(x-1)}=\frac{3\pi^2}{32}$$

(A) $f(x)=\lim\limits_{n\to\infty}\frac{1}{1+n\sin^2\pi x}$

$$=\begin{cases}0 & 0<x<1\\ 1 & x=1\\ 0 & 2>x>1\end{cases}$$

$\therefore$ f(x) is not continuous at x = 1.

2. Given $f(x)=\frac{[\{|x|\}]e^{x^2}\{[x+\{x\}]\}}{\left(e^{1/x^2}-1\right)\text{sgn}(\sin x)}$ for $x\neq0$

$= 0$ for $x = 0$

where {x} is the fractional part function; [x] is the step up function and sgn(x) is the signum function of x then, f(x)

(A) is continuous at x = 0 (B) is discontinuous at x = 0

(C) has a removable discontinuity at x = 0 (D) has an irremovable discontinuity at x = 0

दिया है कि $f(x)=\frac{[\{|x|\}]e^{x^2}\{[x+\{x\}]\}}{\left(e^{1/x^2}-1\right)\text{sgn}(\sin x)}$ for $x\neq0$

$= 0$ for $x = 0$

जहाँ {x} एक अपूर्णांश फलन [x] एक महत्तम पूर्णांक फलन है तथा sgn(x) सिग्नम फलन हो, तो फलन f(x)

(A) x = 0 पर सतत् होगा (B) x = 0 पर असतत्

(C) x = 0 पर विस्थापनीय असतता (D) x = 0 पर अविस्थापनीय असतत्ता

Ans. A

Sol.
$$f(x)=\begin{cases}\dfrac{\left[\{|x|\}\right]e^{x^2}\left\{\left[x+\{x\}\right]\right\}}{\left(e^{\frac{1}{x^2}}-1\right)\text{sgn}(\sin x)} & x\neq0\\ 0 & x=0\end{cases}$$

at $x = 0$

$$LHL = \lim_{h\to 0} f(0-h)$$

$$= \lim_{h\to 0} \frac{[\{h\}]e^{h^2}\{[-h+\{-h\}]\}}{\text{sgn}(-\sinh)\left(e^{\frac{1}{h^2}}-1\right)}$$

$$= \frac{0\cdot e^{h^2}\cdot(1-2h)}{-1\cdot\left(e^{\frac{1}{h^2}}-1\right)} = 0$$

$$RHL = \lim_{h\to 0} \frac{[\{h\}]e^{h^2}\{[h+\{h\}]\}}{\text{sgn}(\sinh)\left(e^{\frac{1}{h^2}}-1\right)} = 0$$

$f(0) = 0$

$\therefore$ f(x) is continuous at $x = 0$.

3. Consider $f(x) = \begin{cases} x[x]^2\log_{(1+x)}2 & \text{for} -1 < x < 0 \\ \dfrac{ln\left(e^{x^2}+2\sqrt{\{x\}}\right)}{\tan\sqrt{x}} & \text{for}\ 0 < x < 1 \end{cases}$

where [ ] & {} are the greatest integer function & fractional part function respectively, then

(A) $f(0) = ln2 \Rightarrow$ f is continuous at $x = 0$ (B) $f(0) = 2 \Rightarrow$ f is continuous at $x = 0$

(C) $f(0) = e^2 \Rightarrow$ f is continuous at $x = 0$ (D) f has an irremovable discontinuity at $x = 0$

दिया है $f(x) = \begin{cases} x[x]^2\log_{(1+x)}2 & \text{for} -1 < x < 0 \\ \dfrac{ln\left(e^{x^2}+2\sqrt{\{x\}}\right)}{\tan\sqrt{x}} & \text{for}\ 0 < x < 1 \end{cases}$

जहाँ [ ] महत्तम पूर्णांक फलन तथा {} भिन्नांश फलन हो, तो

(A) $f(0) = ln2 \Rightarrow x = 0$ पर f सतत है। (B) $f(0) = 2 \Rightarrow x = 0$ पर f सतत है

(C) $f(0) = e^2 \Rightarrow x = 0$ पर f सतत् है। (D) $x = 0$ पर f अविस्थापनीय असतत् है

Ans. D

Sol. $f(x)=\begin{cases} x[x]^2\log_{1+x}2 & -1<x<0 \\ \dfrac{\ln\left(e^{x^2}+2\sqrt{\{x\}}\right)}{\tan\sqrt{x}} & 0<x<1\end{cases}$

$f(x)=\begin{cases} -x\log_{(1+x)}2 & -1<x<0 \\ \dfrac{\ln\left(e^{x^2}+2\sqrt{x}\right)}{\tan\sqrt{x}} & 0<x<1\end{cases}$

at $x=0$

RHL

$=\lim\limits_{h\to 0}\dfrac{\ln\left(e^{h^2}+2\sqrt{h}\right)}{\sqrt{h}}$

4. Consider $f(x)=\dfrac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}}, x\neq 0;$

$g(x)=\cos 2x,\ -\dfrac{\pi}{4}<x<0,$

$h(x)-\begin{cases}\dfrac{1}{\sqrt{2}}f(g(x)) & \text{for } x<0\\ 1 & \text{for } x=0\\ f(x) & \text{for } x>0\end{cases}$

then, which of the following holds good.

where $\{x\}$ denotes fractional part function.

(A) 'h' is continuous at $x=0$

(B) 'h' is discontinuous at $x=0$

(C) $f(g(x))$ is an even function

(D) $f(x)$ is an even function

दिया है $f(x)=\dfrac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}}, x\neq 0;$

$g(x)=\cos 2x,\ -\dfrac{\pi}{4}<x<0,$

$h(x)-\begin{cases}\dfrac{1}{\sqrt{2}}f(g(x)) & \text{for } x<0\\ 1 & \text{for } x=0\\ f(x) & \text{for } x>0\end{cases}$

तो निम्न में से कौन सा सत्य होगा

यहाँ $\{x\}$ भिन्नांश फलन है।

(A) $x = 0$ पर 'h' सतत् है (B) $x = 0$ पर 'h' असतत् है (C) f(g(x)) समफलन है (D) f(x) एक समफलन है

Ans. A

Sol. $f(x)=\dfrac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}};x\neq 0$

$$f(-x)=\frac{\sqrt{1-x}-\sqrt{1+x}}{\{x\}}=\frac{-\left(\sqrt{1+x}-\sqrt{1-x}\right)}{-x-[-x]}$$

$\neq f(x)$

$\therefore$ f(x) is not an even function.

$$h(x)=\begin{cases}\dfrac{1}{\sqrt{2}}f(g(x)) & x<0\\ 1 & x=0\\ f(x) & x>0\end{cases}$$

$$=\begin{cases}\dfrac{1}{\sqrt{2}}f(\cos 2x) & x<0\\ 1 & x=0\\ \dfrac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}} & x>0\end{cases}$$

at $x = 0$

$$\text{LHL}=\lim_{h\to 0}\frac{1}{\sqrt{2}}f(\cos 2h)=\frac{1}{\sqrt{2}}\left(\frac{\sqrt{1+\cos 2h}-\sqrt{1-\cos 2h}}{\{\cos 2h\}}\right)$$

$$=\frac{1}{\sqrt{2}}\sqrt{2}\lim_{h\to 0}\frac{(\cosh-\sinh)}{\cos 2h}=1$$

$$\text{RHL}=\lim_{x\to 0^+}\frac{2x}{\sqrt{1+x}+\sqrt{1-x}}\cdot\frac{1}{\{x\}}=\lim_{h\to 0}\frac{2h}{(h)\left(\sqrt{1-h}+\sqrt{1+h}\right)}=1$$

$\Rightarrow$ h(x) is continuous at $x = 0$.

$$-\frac{\pi}{4}<x<0$$

$$\Rightarrow f(g(x))=f(\cos 2x)$$

$$=\frac{\sqrt{1+\cos 2x}-\sqrt{1-\cos 2x}}{\cos 2x-[\cos 2x]}$$

$$= \frac{\sqrt{2}\left(|\cos 2x| - |\sin x|\right)}{\cos 2x - [\cos 2x]}$$

$$\because x \in \left(-\frac{\pi}{4}, 0\right) \Rightarrow 2x \in \left(-\frac{\pi}{2}, 0\right) \Rightarrow [\cos 2x] = 0$$

$$\Rightarrow f(g(x)) = \frac{\sqrt{2}(\cos x + \sin x)}{\cos 2x}$$

$\because$ f(g(x)) is not an even function.

5. Consider the function $f(x) = \lim\limits_{n \to \infty} \dfrac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$, where $n \in N$

Statement-1: $f(x)$ is discontinuous at $x = 1$. **because**

Statement-2: $f(1) = 0$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true

दिया है कि $f(x) = \lim\limits_{n \to \infty} \dfrac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$, जहाँ $n \in N$

कथन-1: $f(x)$, $x = 1$ पर असतत है।

क्योंकि

कथन-2: $f(1) = 0$.

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है।

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है।

(C) कथन–1 सत्य है, कथन–2 असत्य है।

(D) कथन–1 असत्य है, कथन– 2 सत्य है।

Ans. B

Sol. $$f(x) = \lim_{n \to \infty} \frac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$$

$$= \lim_{n \to \infty} \frac{\dfrac{\sin \pi x}{x^{2n}} - \sin(x-1)}{\dfrac{1}{x^{2n}} + (x-1)}$$

$$= \begin{cases} \sin \pi x & x < 1 \\ 0 & x = 1 \\ -\dfrac{\sin(x-1)}{x-1} & x > 1 \end{cases}$$

$\Rightarrow$ LHL = 0

RHL = –1

$\therefore$ f(x) is not continuous at x = 1.

6. Given $f(x) = \begin{cases} \log_a\left(a\left|[x]+[-x]\right|\right)^x \left(\dfrac{a^{\frac{2}{\left(\frac{[x]+[-x]}{|x|}\right)}-5}}{3+a^{\frac{1}{|x|}}}\right) & \text{for } |x| \neq 0 \text{ ; } a > 1 \\ 0 & \text{for } x = 0 \end{cases}$ where [ ] represents the integral part function, then :

(A) f is continuous but not differentiable at x = 0

(B) f is cont. & diff. at x = 0

(C) the differentiability of 'f' at x = 0 depends on the value of a

(D) f is cont. & diff. at x = 0 and for a = e only

दिया है कि $f(x) = \begin{cases} \log_a\left(a\left|[x]+[-x]\right|\right)^x \left(\dfrac{a^{\frac{2}{\left(\frac{[x]+[-x]}{|x|}\right)}-5}}{3+a^{\frac{1}{|x|}}}\right) & \text{for } |x| \neq 0 \text{ ; } a > 1 \\ 0 & \text{for } x = 0 \end{cases}$ जहाँ [ ] महत्तम पूर्णांक फलन है, तो

(A) x = 0 पर f सतत् होगा लेकिन अवकलनीय नहीं होगा

(B) x = 0 पर f सतत् तथा अवकलनीय दोनों होगा

(C) x = 0 पर 'f' की अवकलनीयता a के मान पर निर्भर करेगी

(D) केवल a = e होने पर ही f, x = 0 पर सतत् तथा अवकलनीय होगा।

Ans. B

Sol. $f(x) = \begin{cases} x.\dfrac{a^2|x|-5}{3+a^{\frac{1}{|x|}}} & x \neq 0 \\ 0 & x = 0 \end{cases} \quad x \neq 0$

$$f(0^-) = \lim_{h\to 0} -h.\frac{a^{2h-5}}{3+a^{\frac{1}{h}}} = 0$$

$$f\left(0^{+}\right)=\lim_{h\to 0}-h.\frac{a^{2h-5}}{3+a^{\frac{1}{h}}}=0$$

f(a) = 0 f is cannot at x = 0

$$\text{L.H.D.} = f'\left(0^{-}\right)=\lim_{h\to 0}-h\frac{\frac{a^{2h-5}}{3+a^{\frac{1}{h}}}}{-h}=\frac{a^{-5}}{3}$$

$$\text{R.H.D.} = f'\left(0^{+}\right)=\lim_{h\to 0}h\frac{\frac{a^{2h-5}}{3+a^{\frac{1}{h}}}}{h}=\frac{a^{-5}}{3}$$

L.H.D. = R.H.D. $\Rightarrow$ f is diff. at x = 0

7. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in I$ and p is a prime number. The number of points where $f(x)$ is not differentiable is

(A) p – 1 (B) p + 1 (C) 2p + 1 (D) 2p – 1.

Here [x] denotes greatest integer function.

माना कि $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in I$ तथा p एक अभाज्य संख्या हो, तो ऐसे कुल कितने बिन्दु होगें जहाँ $f(x)$ अवकलनीय नहीं होगा।

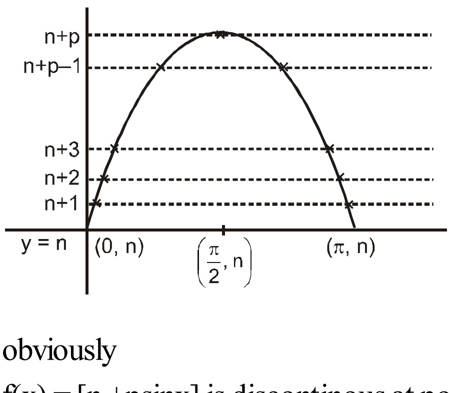
(A) p – 1 (B) p + 1 (C) 2p + 1 (D) 2p – 1.

यहाँ [.] महत्तमपूर्णांक फलन है।

Ans. D

Sol. $f(x) = [n + p\sin x]$, $x \in (0, \pi)$

graph of $y = n + p \sin x$



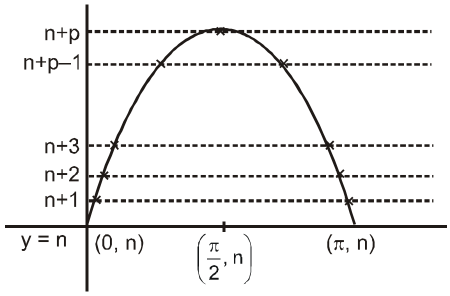
obviously

f(x) = [n +psinx] is discontinous at points mark in above curve

$\Rightarrow$ number of such points $\Rightarrow$ (p –1) + 1 + p –1 = 2p –1

Hindi. $f(x) = [n + p\sin x]$, $x \in (0, \pi)$

$y = n + p \sin x$ का आलेख



स्पष्ट रूप से

f(x) = [n +psinx] असतत् है उन बिन्दुओं पर जो आलेख से चिन्हित है।

$\Rightarrow$ ऐसे बिन्दुओं की संख्या $\Rightarrow$ $(p-1) + 1 + p - 1 = 2p - 1$

8. Consider $f(x) = \left[\dfrac{2\left(\sin x - \sin^3 x\right) + \left|\sin x - \sin^3 x\right|}{2\left(\sin x - \sin^3 x\right) - \left|\sin x - \sin^3 x\right|}\right]$, $x \neq \dfrac{\pi}{2}$ for $x \in (0, \pi)$

$f(\pi/2) = 3$ where [ ] denotes the greatest integer function then,

(A) f is continuous & differentiable at $x = \pi/2$

(B) f is continuous but not differentiable at $x = \pi/2$

(C) f is neither continuous nor differentiable at $x = \pi/2$

(D) none of these

दिया है कि $f(x) = \left[\dfrac{2\left(\sin x - \sin^3 x\right) + \left|\sin x - \sin^3 x\right|}{2\left(\sin x - \sin^3 x\right) - \left|\sin x - \sin^3 x\right|}\right]$, $x \neq \dfrac{\pi}{2}$ $x \in (0, \pi)$

$f(\pi/2) = 3$ जहाँ [ ] महत्तम पूर्णांक फलन हो, तो

(A) $x = \pi/2$ पर f सतत् तथा अवकलनीय दोनों होगें।

(B) $x = \pi/2$ पर f सतत् तो होगा लेकिन अवकलनीय नहीं है।

(C) $x = \pi/2$ पर f न तो सतत् हो और न ही अवकलनीय है।

(D) इनमें से कोई नहीं

Ans. A

Sol. In the immediate neighborhood of $x = \pi/2$, $\sin x > \sin^3 x \Rightarrow |\sin x - \sin^3 x| = \sin x - \sin^3 x$

Hence for $x \neq \pi/2$, $f(x) = \left[\dfrac{2(\sin x - \sin^3 x) + \sin x - \sin^3 x}{2(\sin x - \sin^3 x) - \sin x + \sin^3 x}\right] = \dfrac{3\sin x - 3\sin^3 x}{\sin x - \sin^3 x} = 3$

Hence f is continuous and diff. at $x = \pi/2$]

9. Let $f : [a, b] \to R$ be any function which is such that $f(x)$ is rational for irrational x and that $f(x)$ is irrational for rational x, then in [a, b]

(A) *f* is discontinuous everywhere in [a, b]

(B) *f* is continuous only at x = 0 and discontinuous everywhere

(C) f is continuous for all irrational x and discontinuous for rational x

(D) f is continuous for rational x and discontinuous for irrational x

$f : [a, b] \to R$ एक ऐसा फलन है कि यदि $x \notin Q$ तो $f(x)$ परिमेय है तथा $x \in Q$ तो $f(x)$ अपरिमेय हो, तो अन्तराल [a, b] में

(A) [a, b] सभी जगह f असतत् होगा

(B) फलन f सिर्फ $x = 0$ पर सतत् होगा बाकी बिन्दुओं पर असतत् होगा।

(C) यदि x अपरिमेय हो, तो f सतत् होगा तथा यदि x परिमेय हो, तो f असतत् होगा।

(D) यदि x परिमेय हो तो f सतत् होगा तथा यदि x अपरिमेय हो, तो f असतत् होगा।

Ans. A

Sol. $$f(x) = \begin{cases} \text{rational} & \text{if } x \notin Q \text{ in } [a,b] \\ \text{irrational} & \text{if } x \in Q \text{ in } [a,b] \end{cases}$$

Let $c \in Q$

$$\left.\begin{array}{l} f(c) = \text{irrational} \\ f(c+h) = \text{rational} \\ \lim\limits_{h\to 0} \text{ through irrational} \end{array}\right] \Rightarrow f \text{ is discontinuous for } x \in Q \text{ in } [a, b]$$

now let $r \notin Q$

$$\left.\begin{array}{l} f(r) = \text{rational} \\ f(r+h) = \text{irrational} \\ \lim\limits_{h\to 0} \text{ through rational} \end{array}\right] \Rightarrow f \text{ is discontinuous for } x \notin Q \text{ in } [a, b]$$

10. The graph of function f contains the point P (1, 2) and Q(s, r). The equation of the secant line through P and Q is $y = \left(\dfrac{s^2+2s-3}{s-1}\right)x - 1 - s$. The value of $f'(1)$, is

(A) 2 (B) 3 (C) 4 (D) non existent

फलन $f(x)$ का आरेख बिन्दु P (1, 2) तथा Q(s, r) से गुजरता है यदि P और Q से गुजरने वाली रेखा खण्ड का समीकरण $y = \left(\dfrac{s^2+2s-3}{s-1}\right)x - 1 - s$ हो, तो $f'(1)$ का मान बताओं।

(A) 2 (B) 3 (C) 4 (D) अस्तित्व विहीन

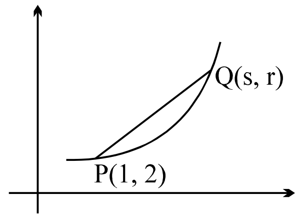
Ans. C

Sol. **I** By definition $f'(1)$ is the limit of the slope of the secant line when $s \to 1$.

$$\text{Thus } f'(1) = \lim_{s \to 1} \frac{s^2 + 2s - 3}{s - 1}$$

$$= \lim_{s \to 1} \frac{(s-1)(s+3)}{s-1}$$

$$= \lim_{s \to 1} (s+3) = 4 \qquad \Rightarrow \qquad (D)$$



II By substituting x = s into the equation of the secant line, and cancelling by s – 1 again, we get $y = s^2 + 2s - 1$. This is f (s), and its derivative is $f'(s) = 2s + 2$, so $f'(1) = 4$.]

11. If $f(x + y) = f(x) + f(y) + |x|y + xy^2$, $\forall$ x, y $\in$ R and $f'(0) = 0$, then

(A) f need not be differentiable at every non zero x (B) f is differentiable for all $x \in R$

(C) f is twice differentiable at x = 0 (D) none

यदि $f(x + y) = f(x) + f(y) + |x|y + xy^2$, $\forall$ x, y $\in$ R और $f'(0) = 0$, तो

(A) कोई जरूरी नहीं की प्रत्येक अशून्य (zero) के लिये f अवकलनीय हो

(B) x के प्रत्येक मान के लिये f एक अवकलनीय फलन होगा

(C) x = 0 पर f तथा f' दोनो अवकलनीय होगा

(D) इनमें से कोई नहीं

Ans. B

Sol. $$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{f(h) + |x|h + xh^2}{h} \quad \text{where } x = h \text{ and } y = x$$

$$\because \ f(0) = 0; \quad \text{hence } f'(x) = \lim_{h \to 0} \left(\frac{f(h) - f(0)}{h} + |x| + xh \right)$$

$$f'(x) = f'(0) + |x| = |x|$$

12. For x > 0, let $h(x) = \begin{cases} \frac{1}{q} & \text{if } x = \frac{p}{q} \\ 0 & \text{if } x \text{ is irrational} \end{cases}$ *where p & q > 0 are relatively prime integers*

then which one does not hold good?

(A) h(x) is discontinuous for all x in $(0, \infty)$

(B) h(x) is continuous for each irrational in $(0, \infty)$

(C) h(x) is discontinuous for each rational in $(0, \infty)$

(D) h(x) is not derivable for all x in $(0, \infty)$

x > 0 के लिये $h(x) = \begin{cases} \frac{1}{q} & \text{यदि } x = \frac{p}{q} \\ 0 & \text{यदि } x \text{ अपरिमेय है} \end{cases}$

जहाँ *p, q > 0* तथा p और q तुलनात्मक अविभाज्य संख्याऐं हो, तो इनमें से कौन सा विकल्प असत्य होगा?

(A) $(0, \infty)$ के अन्तराल में सभी x के लिए h(x) असतत् फलन होगा

(B) $(0, \infty)$ के अन्तराल में प्रत्येक अपरिमेय के लिये h(x) सतत् फलन है।

(C) (0, ∞) के अन्तराल में प्रत्येक परिमेय के लिये h(x) असतत् फलन है।

(D) (0, ∞) के अन्तराल में x के किसी भी मान के लिये h(x) अवकलनीय् फलन नहीं होगा

Ans. A

Sol. Let $x = \frac{2}{3}$ which is rational

$\Rightarrow \quad h\left(\frac{2}{3}\right) = \frac{1}{3}$

$$\lim_{t \to 0} h\left(\frac{2}{3} + t\right) = 0 \quad \Rightarrow \text{discontinuous at } x \in Q$$

Let $x = \sqrt{2} \notin Q$

$h\left(\sqrt{2}\right) = 0$ consider $\sqrt{2} = 1.41401235839$

$$h\left(\sqrt{2}\right) = h\left(\frac{1.414023583}{10^{10}}\right) = \frac{1}{10^{10}} \to 0$$

Hence h is continuous for all irrational $\Rightarrow$ A

13. Let $f(x) = \max.\left\{\left|x^2 - 2|x|\right|, |x|\right\}$ and $g(x) = \min.\left\{\left|x^2 - 2|x|\right|, |x|\right\}$ then

(A) both f (x) and g (x) are non differentiable at 5 points.

(B) f (x) is not differentiable at 5 points whether g (x) is non differentiable at 7 points.

(C) number of points of non differentibility for f (x) and g (x) are 7 and 5 respectively.

(D) both f (x) and g (x) are non differentiable at 3 and 5 points respectively

माना $f(x) = \max.\left\{\left|x^2 - 2|x|\right|, |x|\right\}$ तथा $g(x) = \min.\left\{\left|x^2 - 2|x|\right|, |x|\right\}$ है तो –

(A) f (x) तथा g (x) दोनों 5 बिन्दुओं पर अवकलनीय नहीं है

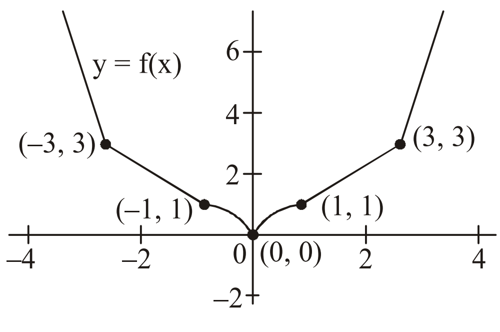
(B) f (x), 5 बिन्दुओं पर अवकलनीय नहीं है जबकि g (x), 7 बिन्दुओं पर अवकलनीय नहीं है

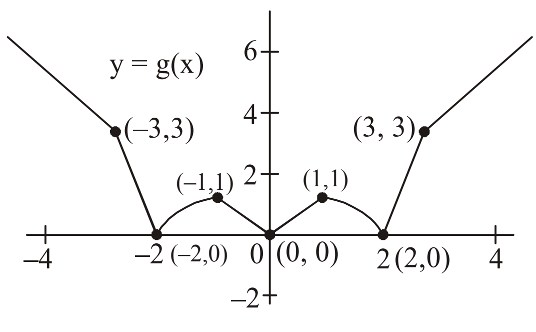
(C) f (x) तथा g (x) क्रमशः 7 तथा 5 बिन्दुओं पर अवकलनीय नहीं है

(D) f (x) तथा g (x) क्रमशः 3 तथा 5 बिन्दुओं पर अवकलनीय नहीं है

Ans. B

Sol. 0





ONE OR MORE THAN ONE CORRECT TYPE

14. If f is defined on an interval [a, b]. Which of the following statement(s) is/are INCORRECT?

(A) If f(a) and f(b), have opposite sign, then there must be a point c $\in$ (a, b) such that $f(c) = 0$.

(B) If f is continuous on [a, b], $f(a) < 0$ and $f(b) > 0$, then there must be a point c $\in$ (a, b) such that $f(c) = 0$

(C) If f is continuous on [a, b] and there is a point c in (a, b) such that $f(c) = 0$, then f(a) and f(b) have opposite sign.

(D) If f has no zeroes on [a, b], then f(a) and f(b) have the same sign

फलन f अन्तराल [a, b] में परिभाषित हो, तो गलत विकल्पों को चुनिये।

(A) यदि f(a) और f(b) का चिन्ह विपरीत हो, तो एक ऐसा c $\in$ (a, b) होगा जब $f(c) = 0$.

(B) यदि f अन्तराल [a, b] में सतत् है तथा $f(a) < 0$ और $f(b) > 0$ हो, तो एक ऐसा c $\in$ (a, b) होगा जब $f(c) = 0$

(C) यदि f अन्तराल [a, b] में सतत् है तथा c $\in$ (a, b) में एक ऐसा बिन्दु है जब $f(c) = 0$ तो f(a) और f(b) का चिन्ह विपरीत होगा।

(D) यदि [a, b] में f का कोई मूल नही है तो f(a) व f(b) का चिन्ह समान होगा

Ans. ACD

Sol. According to intermediate value theorem if f(a) f(b) < 0 then there exist at least one 'c' $\in$ (a, b) for which f(x) = 0.

15. If $f(x) = \text{Min}(\tan x, \cot x)$ then:

(A) f(x) is discontinuous at $x = 0, \frac{\pi}{4}$ & $\frac{5\pi}{4}$ (B) f(x) is continuous at $x = \frac{\pi}{2}$ & $\frac{3\pi}{2}$

(C) $\int_0^{\pi/2} f(x)\,dx = 2\,\ell n\sqrt{2}$ (D) f(x) is periodic with period π

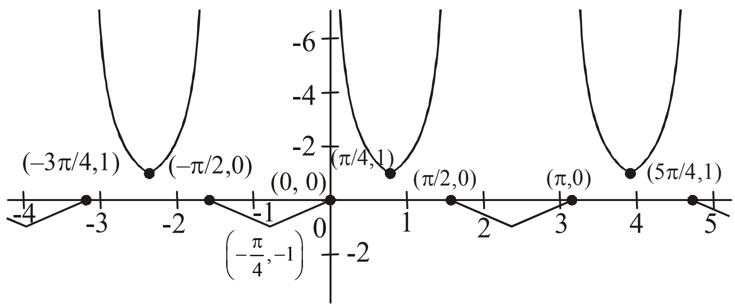
यदि $f(x) = \text{Min}(\tan x, \cot x)$ तो :

(A) f(x) $x = 0, \frac{\pi}{4}$ तथा $\frac{5\pi}{4}$ पर असतत् है (B) फलन f(x) $x = \frac{\pi}{2}$ तथा $\frac{3\pi}{2}$ पर सतत् है

(C) $\int_0^{\pi/2} f(x)\,dx = 2\,\ell n\sqrt{2}$ (D) f(x) का आवर्तकाल π है

Ans. CD

Sol.



(C) $\int_0^{\frac{\pi}{2}} f(x)\,dx = \int_0^{\frac{\pi}{4}} \cot x\,dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \tan x\,dx$

$= 2\ln\sqrt{2}$

(D) Time period = π

16. If $f(x) = |2x+1| + |2x-\pi| + \tan x$ then $f(x)$ is

(A) Continuous at all points except $x = n\pi + \frac{\pi}{2}$ (where $n \in I$)

(B) Continuous at $x = -\frac{1}{2}$, but not differentiable at $x = \pi/2$

(C) Not differentiable at $x = n\pi + \frac{\pi}{2}$ (where $n \in I$)

(D) Limit at $x = \frac{\pi}{2}$ does not exist

यदि $f(x) = |2x+1| + |2x-\pi| + \tan x$ तो फलन $f(x)$ है –

(A) $x = n\pi + \frac{\pi}{2}$ (जहाँ $n \in I$) के अलावा सभी बिन्दुओं पर संतत

(B) $x = -\frac{1}{2}$ पर संतत परन्तु $x = \pi/2$ पर अवकलनीय नहीं

(C) $x = n\pi + \frac{\pi}{2}$ (जहाँ $n \in I$) अवकलनीय नहीं है

(D) $x = \frac{\pi}{2}$ पर सीमा विद्यमान नहीं है

Ans. ABCD

Sol.
$$f(x) = \begin{cases} -4x+\pi-1+\tan x & x < -\frac{1}{2} \\ \pi+1+\tan x & -\frac{1}{2} \le x < \frac{\pi}{2} \\ 4x-\pi+1+\tan x & x > \frac{\pi}{2} \end{cases}$$

$x = \frac{\pi}{2}$

$f\left(\frac{\pi}{2}^{-}\right) \to \infty$

$f\left(\frac{\pi}{2}^{+}\right) \to -\infty$

$x = -\frac{1}{2}$

$$f\left(-\frac{1}{2}^{-}\right) = -4\left(-\frac{1}{2}\right) + \pi - 1 + \tan\left(-\frac{1}{2}\right) = \pi + 1 - \tan\frac{1}{2}$$

$$f\left(-\frac{1}{2}^{+}\right) = \pi + 1 + \tan\left(-\frac{1}{2}\right) = \pi + 1 - \tan\frac{1}{2}$$

$$LHL = RHL = f\left(\frac{1}{2}\right)$$

17. The function defined as $f(x) = \lim_{n\to\infty} \begin{bmatrix} \cos^{2n} x & \text{if } x < 0 \\ \sqrt[n]{1+x^n} & \text{if } 0 \le x \le 1 \\ \dfrac{1}{1+x^n} & \text{if } x > 1 \end{bmatrix}$. Which of the following does not hold good?

(A) continuous at x = 0 but discontinuous at x = 1.
(B) continuous at x = 1 but discontinuous at x = 0.
(C) continuous both at x = 1 and x = 0.
(D) discontinuous both at x = 1 and x = 0

दिया है $f(x) = \lim_{n\to\infty} \begin{bmatrix} \cos^{2n} x & \text{if } x < 0 \\ \sqrt[n]{1+x^n} & \text{if } 0 \le x \le 1 \\ \dfrac{1}{1+x^n} & \text{if } x > 1 \end{bmatrix}$. तो इनमें से कौन सा विकल्प फलन f(x) के लिये सत्य नहीं होगा?

(A) x = 0 पर सतत् तथा x = 1 पर असतत् (B) x = 1 पर सतत् तथा x = 0 पर असतत्
(C) x = 0 , x = 1 पर सतत् (D) x = 0, x = 1 पर असतत्

Ans. ABC

Sol. $f(0^-) = 0$; $f(0^+) = 1$; $f(0) = 1$ $\Rightarrow$ discontinuous at x = 0
$f(1^-) = 1$; $f(1) = 1$; $f(1^+) = 0$ $\Rightarrow$ discontinuous at x = 1

18. f is a continous function in [a, b]; g is a continuous functin in [b, c]
A function h (x) is defined as
$h(x) = f(x)$ for $x \in [a, b)$
$= g(x)$ for $x \in (b, c]$
if $f(b) = g(b)$, then
(A) h(x) has a removable discontinuity at x=b.
(B) h(x) may or may not be continuous in [a, c]
(C) $h(b^-) = g(b^+)$ and $h(b^+) = f(b^-)$
(D) $h(b^+) = g(b^-)$ and $h(b^-) = f(b^+)$

फलन f [a, b] में सतत् है, फलन g [b, c] में सतत् है। एक फलन h (x) इस प्रकार से परिभाषित किया जाता है कि

$$h(x) = f(x) \quad \text{for } x \in [a, b)$$
$$= g(x) \quad \text{for } x \in (b, c]$$

यदि f (b) = g (b), तो

(A) h(x), x = b पर विस्थापनीय असतता

(B) [a, c] में h(x) सतत हो भी सकता है, और नहीं भी

(C) $h(b^-) = g(b^+)$ & $h(b^+) = f(b^-)$

(D) $h(b^+) = g(b^-)$ & $h(b^-) = f(b^+)$

Ans. AC

Sol. Given f is continuous in [a, b] ....(1)

g is continuous in [b, c] ....(2)

f (b) = g(b) ....(3)

$$\left.\begin{aligned} h(x) &= f(x) && \text{for } x \in [a, b) \\ &= f(b) = g(b) && \text{for } x = b \\ &= g(x) && \text{for } x \in (b, c] \end{aligned}\right\} \quad ....(4)$$

h (x) is continuous in $[a, b) \cup (b, c]$ [using (1), (2)]

also $f(b^-) = f(b)$; $g(b^+) = g(b)$ ....(5) [using (1), (2)]

$\therefore$ $h(b^-) = f(b^-) = f(b) = g(b) = g(b^+) = h(b^+)$ [using (4), (5)

now, verify each alternative. Of course! $g(b^-)$ and $f(b^+)$ are undefined.

$h(b^-) = f(b^-) = f(b) = g(b) = g(b^+)$

and $h(b^+) = g(b^+) = g(b) = f(b) = f(b^-)$

hence $h(b^-) = h(b^+) = f(b) = g(b)$

and h (b) is not defined $\Rightarrow$ (A)]

19. Let f be a continuous function on the closed interval [a, b]. Then which of the following statement(s) need not be true?

(A) There is a number c in the open interval (a, b) such that f(a) < f(c) < f(b).

(B) There is a number c in the closed interval [a, b] such that $f(c) \geq f(x)$ for all x in [a, b].

(C) There is a number c in the open interval (a, b) such that $f'(c) = 0$

(D) There is a number c in the open interval (a, b) such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$

माना f एक सतत् फलन है अन्तराल [a, b] मे, तो निम्न में से कौनसा कथन सही होना आवश्यक नही है ?

(A) अन्तराल (a, b) में एक संख्या c है जिसके लिये f(a) < f(c) < f(b) है

(B) अन्तराल [a, b] में एक संख्या c है जिसके लिये $f(c) \geq f(x)$ सभी $x \in [a, b]$ के लिये

(C) अन्तराल (a, b) में एक संख्या c है जिसके लिये $f'(c) = 0$ है

(D) अन्तराल (a, b) में एक संख्या c है जिसके लिये $f'(c) = \dfrac{f(b) - f(a)}{b - a}$ है

Ans. ACD

Sol. (A) Can be true only if function is strictly increasing in (a, b).

(B) Correct because by extreme value theorem for continuous function.

(C) False. Why $f'(c) = 0$?

(D) Can be true if differentiable in (a, b)].

20. Let $f(x) = \begin{cases} x^2 \cos\dfrac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin\dfrac{1}{x}, & x > 0 \end{cases}$,

then which of the following is(are) correct?

(A) f(x) is continuous but not differentiable at x = 0.

(B) f(x) is continuous and differentiable at x = 0.

(C) $f'(x)$ is continuous but not differentiable at x = 0.

(D) $f'(x)$ is discontinuous at x = 0

माना $f(x) = \begin{cases} x^2 \cos\dfrac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin\dfrac{1}{x}, & x > 0 \end{cases}$, तो निम्न में से कौनसा सही है ?

(A) f(x), x = 0 सतत् है परन्तु अवकलनीय नही है

(B) f(x) x = 0 सतत् है तथा अवकलनीय है

(C) $f'(x)$ x = 0 सतत् है परन्तु अवकलनीय नही है

(D) $f'(x)$, x = 0 असतत् है

Ans. BD

Sol. Clearly f(x) is continuous and differentiable at x = 0 such that $f'(0) = 0$

$$f'(x) = \begin{cases} \sin\dfrac{1}{x} + 2x\cos\dfrac{1}{x}, & x < 0 \\ 0, & x = 0 \\ 2x\sin\dfrac{1}{x} - \cos\dfrac{1}{x}, & x > 0 \end{cases}$$

$\therefore$ Clearly $f'(x)$ is discontinuous at x = 0

21. If $f(x) = \min.(1, x^2, x^3)$, then

(A) f(x) is continuous $\forall\, x \in R$

(B) $f'(x) > 0$, $\forall\, x > 1$

(C) f(x) is not differentiable but continuous $\forall\, x \in R$

(D) f(x) is not differentiable for two values of x

फलन $f(x) = \min.(1, x^2, x^3)$, के लिये

(A) फलन f (x), $\forall x \in R$ के लिये सतत् होगा

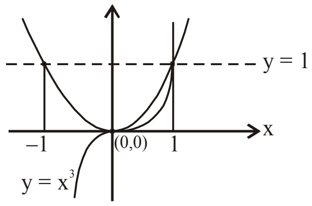
(B) $f'(x) > 0, \forall x > 1$

(C) f(x), $\forall x \in R$ के लिए सतत् तो होगा लेकिन अवकलनीय नही होगा।

(D) f(x), x के दो मान के लिये अवकलनीय नहीं होगा।

Ans. AC

Sol. $f(x) = \min(1, x^2, x^3)$



$$f(x) = \begin{cases} x^3 & x < 0 \\ x^3 & 0 \le x < 1 \\ 1 & x \ge 1 \end{cases}$$

$$f(x) = \begin{cases} x^3 & x < 1 \\ 1 & \ge 1 \end{cases}$$

$\therefore$ f(x) is continuous $\forall x \in R$

f(x) is not differentiable at x = 1.

22. $$f(x) = \begin{cases} 2(x+1) & ; \quad x \le -1 \\ \sqrt{1-x^2} & ; \quad -1 < x < 1 \\ |||x|-1|-1| & ; \quad x \ge 1 \end{cases}, \text{then}$$

(A) f(x) is non differentiable at exactly three points

(B) f(x) is continuous in $(-\infty, 1]$

(C) f(x) is differentiable in $(-\infty, -1)$

(D) f(x) is finite type of discontinuity at x = 1, but continuous at x = –1

$$f(x) = \begin{cases} 2(x+1) & ; \quad x \le -1 \\ \sqrt{1-x^2} & ; \quad -1 < x < 1 \\ |||x|-1|-1| & ; \quad x \ge 1 \end{cases} \text{ है तो –}$$

(A) f(x), ठीक तीन बिन्दुओं पर अवकलनीय नहीं है

(B) $(-\infty, 1]$ में f(x) सतत है

(C) $(-\infty, -1)$ में f(x) अवकलनीय है

(D) x = 1 पर f(x) की विस्थापनीय असतता है लेकिन x = –1 पर सतत है

Ans. ACD

Sol. $$f(x)=\begin{cases} 2(x+1) & x\le -1 \\ \sqrt{1-x^2} & -1<x<1 \\ \left||x-1|-1\right| & x\ge 1 \end{cases}$$

$$f(x)=\begin{cases} 2(x+1) & x\le 1 \\ \sqrt{1-x^2} & -1<x<1 \\ -(x-2) & 1\le x<2 \\ x-2 & 2\le x<3 \end{cases}$$

If $x \in (-\infty, 1)$

$$f'(x)=\begin{cases} 2 & x<-1 \\ \dfrac{-2x}{2\sqrt{1-x^2}} & -1<x<1 \end{cases}$$

$\Rightarrow$ f(x) is not differentiable at $x = -1$.

$\therefore$ f(x) is differentiable $\forall\ x \in (-\infty,-1)$.

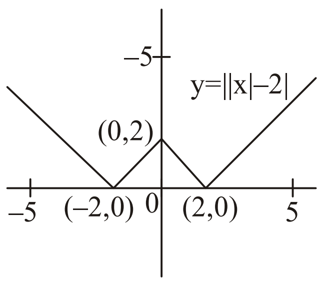
23. If $f(x) = ||\,|x| - 2| + p|$ have more than 3 points of non-derivability then the value of p can be

दि $f(x) = ||\,|x| - 2| + p|$, 3 से अधिक बिन्दुओं पर अवकलनीय नहीं है तो p का मान हो सकता है

(A) 0 (B) –1 (C) –2 (D) 2

Ans. BC

Sol.



Clearly, for p = 0 $f(x) = ||\,|x| - 2| + p|$ will be same as the above graph, and it is non differentiable for exactly 3 points.

Now, for more than 3 points of non diffferentiability, we need to shift the graph down and take the modulus.

$\therefore$ We need to shift the graph of $||x| - 2|$ down. So, p has to be negative. i.e. $p < 0$